Features At A Glance

- Based on ESP32-S3 microcontroller with Xtensa 32-bit LX7 dual-core processor, capable of running at 240 MHz
- Integrated 2.4GHz Wi-Fi and Bluetooth 5 (LE) dual-mode wireless communication, with superior RF performance
- Onboard isolated RS485 interface, for connecting to various RS485 Modbus industrial modules or sensors
- Onboard isolated CAN interface for easy access to various CAN devices
- Onboard power isolation for connecting external devices
- Onboard USB Type-C port for power supply, firmware downloading and debugging
- Onboard power supply screw terminal, supports 7~36V wide voltage input, suitable for industrial applications
- Onboard RTC chip, supports scheduled tasks
- Onboard digital isolation to prevent interference from external signal
- Onboard unitively power supply isolation, providing stable isolated voltage, no extra power supply is required for the isolated terminal
- Equipped RS485 TX/RX indicators and CAN indicator for monitoring the operating status of the module
- Fast-mount protective case, easy to install, safe to use

Specifications

MICROCONTROLLER		ESP32-S3 (Default module: ESP32-S3R8, customizable for other modules)
WIRELESS COMMUNICATION		2.4GHz Wi-Fi (802.11 b/g/n), Bluetooth 5 (LE)
USB	Connector	USB Type-C
	Power supply	5V
ISOLATED RS485	Functions	Power supply, USB communication, firmware downloading, etc.
	Connector	Screw terminal
	Isolation power	Automatically controlled via main controller hardware-free settings
	Protection	TVS diode, surge protection & ESD protection
	Rounter	Onboard internal 120Ω matching resistor, 1k½, default: disabled, no jumper
ISOLATED CAN	Connector	Screw terminal
	Standard control	Hardware automatic control
	Protection	TVS diode, surge protection & ESD protection
	Rounter	Onboard internal 120Ω matching resistor, 1k½, default: disabled, no jumper
LED INDICATORS	USB indicator	Red power indicator, lights up when there is USB connection and voltage is detected
	RS485 indicator	lights up green when the RS485 port sends data lights up blue when the RS485 port receives data
	CAN indicator	Blinks during data transmission
POWER SUPPLY SCREW TERMINAL	Voltage range	7V ~ 36V
	Endurance	Fast-mount protective case
APPEARANCE	Dimensions	91.4×72.5×34.7 (mm)

Based On ESP32-S3, Designed For AIoT Market

Equipped With Xtensa 32-Bit LX7 Dual-Core Processor, Capable Of Running At 240 MHz, With Powerful AI Computing Performance And Security Encryption Mechanism, Integrated 2.4GHz Wi-Fi And Bluetooth 5 (LE) Dual-Mode Wireless Communication

Onboard Multiple Isolation Protection Circuit

Multiple Protections, More Safe And Reliable

Application Scenario

Allows Users To Access The Relevant Webpage Via A Mobile Phone Or PC Browser To Control The Device And Send Data

Supports Arduino Development

Comprehensive SDK, Dev Resources, And Tutorials To Help You Easily Get Started

Arduino IDE

Arduino IDE is an open source electronic prototyping platform, convenient and flexible, easy to get started.

Application Examples

Provides Demo Code For IoT Connectivity By Integrating Communication Interfaces With A Web Page, Enabling Data Visualization Services

Interface Description

TOP TERMINAL BLOCK DESCRIPTION			BOTTOM TERMINAL BLOCK DESCRIPTION		
POWER SUPPLY TERMINAL	Power positive (7V ~ 36V)		RS485 TERMINAL	5V signal pin	
	Power negative			5V signal pin	
TYPE-C	Type-C power supply, firmware download, and communication			A signal pin	
	GND	Ground (GND)		B signal pin	
SHI-2	5V	5.0V power output	CAN TERMINAL		
	CPO2	ESP32-S3 IO pin			
	CPI2	ESP32-S3 IO pin			

What's On Board

1. ESP32-S3
2. Wi-Fi antenna (frequency band: 2.4GHz Wi-Fi and BLE)
3. 4M qSPI pseudo static RAM (PSRAM)
4. Pin header (2.54mm pitch)
5. Power supply screw terminal (isolation)
6. Power supply screw terminal (non-isolated) (supports 7V ~ 36V wide voltage input)
7. Type-C port (USB)
8. 7~36V power supply, firmware download and USB communication
9. RTC battery holder
10. RTC battery (battery specification: CR1220, battery size: 12 × 2mm, nominal capacity: 40mAh)
11. SPI2 + Ethernet
12. Power supply isolation, isolated DC-DC module (5V, 1W)
13. Isolated power supply
14. Digital isolation
15. Isolated power supply (non-isolated signal)
16. Isolated power supply
17. USB indicator
18. Power supply indicator
19. RS485 signal power pins (for RS485 port sends data, lights up blue when the RS485 port receives data)
20. CAN signal power pins (for CAN port sends data, lights up blue when the CAN port receives data)
21. RS485 TX/RX indicator
22. SHI2 RS485 Interface
23. Power supply isolation, isolated DC-DC module (5V, 1W)
24. Isolated power supply (non-isolated signal) (Note: 5V power supply is required for the isolated terminal)
25. Isolated power supply
26. Isolated power supply (non-isolated signal)
27. Isolated power supply
28. Isolated power supply (non-isolated signal)
29. Isolated power supply
30. Isolated power supply (non-isolated signal)
31. Isolated power supply
32. Isolated power supply (non-isolated signal)
33. Isolated power supply
34. Isolated power supply (non-isolated signal)
35. Isolated power supply
36. Isolated power supply (non-isolated signal)
37. Isolated power supply
38. Isolated power supply (non-isolated signal)
39. Isolated power supply
40. Isolated power supply (non-isolated signal)

Enclosure Design

Protection Enclosure With Rail-Mount Support, Easy To Install, Safe To Use

Outline Dimensions

Support Batch Customization

Support Software And Hardware Customization
Including Hardware Interface, LOGO, Label, Case And Web Page, Etc.

Label Customization

Hardware Customization

Software Customization

Contact customer service for your customization needs >>>> One-to-one professional and technical service for you >>>> Signing of R&D/sales customization contract and start project customization >>>> Batch production will be delivered after the sample product is passed

Resources & Services

WAVESHARE ORIGINAL

PROFESSIONAL TECH SUPPORT

RICH RESOURCES MANUAL & DEMO

Wiki: www.waveshare.com/wiki/ESP32-S3-RS485-CAN

Selection Guide

MODEL	DI	DO	RELAY	RS485	CAN	NETWORK PORT	POE	RTC	BUZZER	TF	ANTENNA	PIN HEADER
ESP32-S3-Relay-6CH	-	-	6	√	-	-	-	-	√	-	√	40PIN header, compatible with Raspberry Pi Pico interface, 2.54mm pitch
ESP32-S3-ETH-8DI-8RO	8	-	8	√	-	√	-	√	√	√	√	28PIN header, 2.54mm pitch
ESP32-S3-POE-ETH-8DI-8RO	8	-	8	√	-	√	√	√	√	√	√	28PIN header, 2.54mm pitch
ESP32-S3-ETH-8DI-8RO-C	8	-	8	-	√	√	-	√	√	√	√	28PIN header, 2.54mm pitch
ESP32-S3-POE-ETH-8DI-8RO-C	8	-	8	-	√	√	√	√	√	√	√	28PIN header, 2.54mm pitch
ESP32-S3-POE-ETH-8DI-8DO	8	8	-	√	√	√	√	√	√	√	√	8PIN screw terminal, 5.0mm pitch
ESP32-S3-Relay-1CH	-	-	1	√	-	-	-	√	-	-	√	20PIN header, 2.54mm pitch
ESP32-S3-RS485-CAN	-	-	-	√	√	-	-	√	-	-	√	20PIN header, 2.54mm pitch

15 Days Returnable
12 Months Warranty
Trustworthy Brand
Discount for Large Quantity
Worldwide Shipping

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